

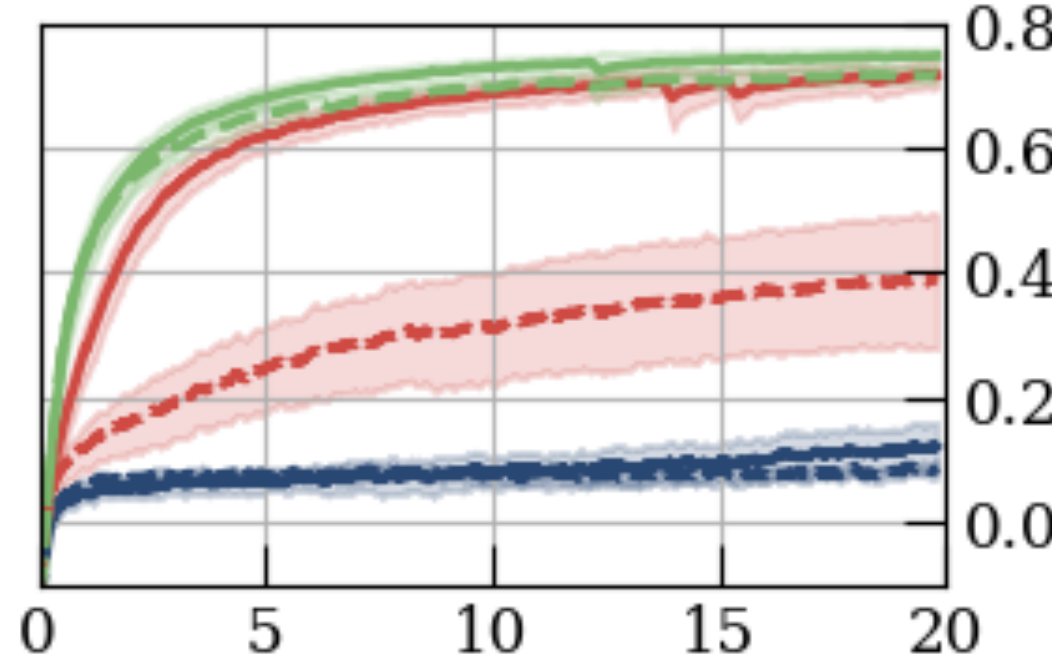
• Improve sample efficiency in TC-RL and TC-MARL.

RAP Embeddings

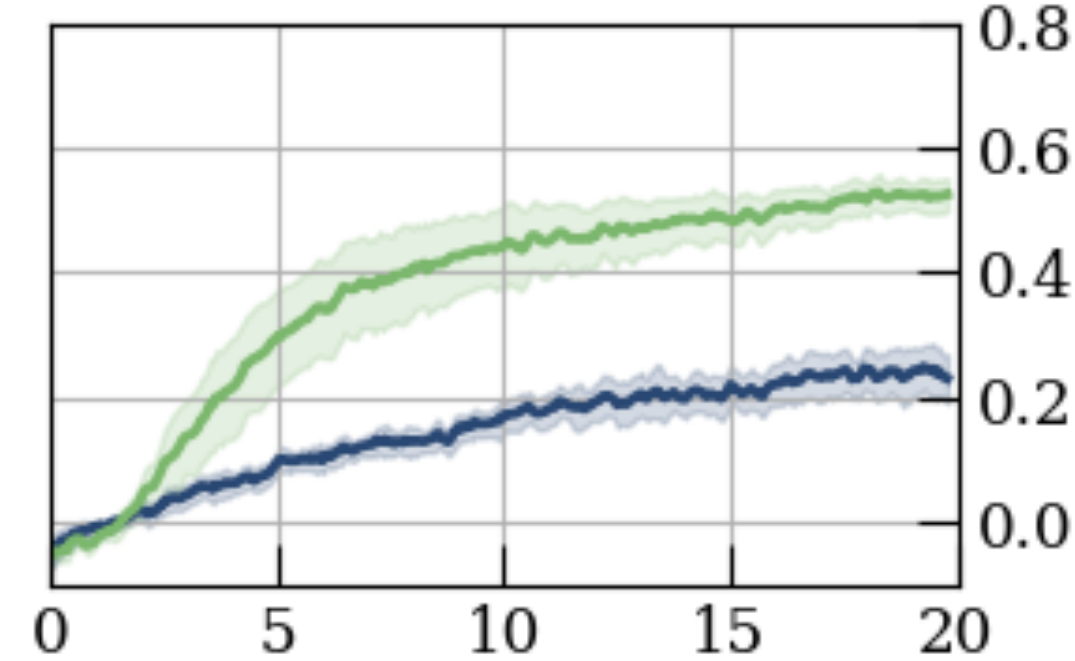
Automata-Conditioned RL

Discounted return

Letterworld



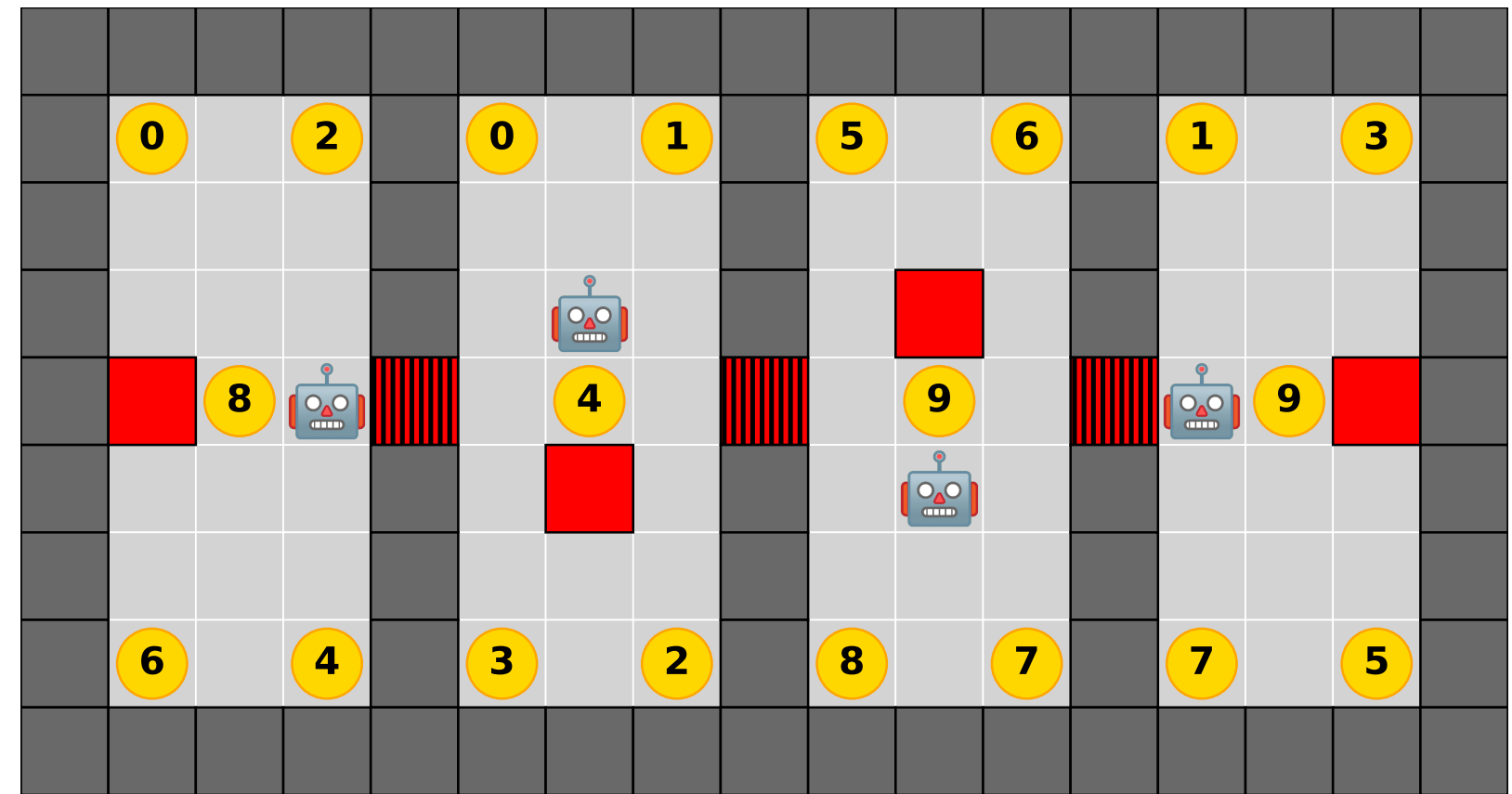
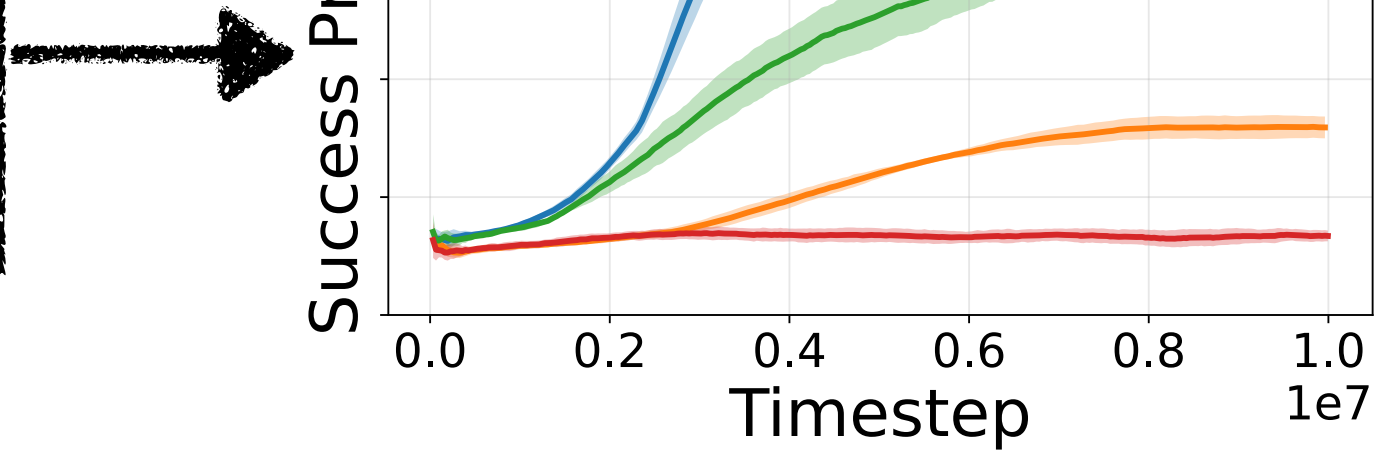
Zones



Frames (millions)

Frames (millions)

Automata-Conditioned Cooperative MARL





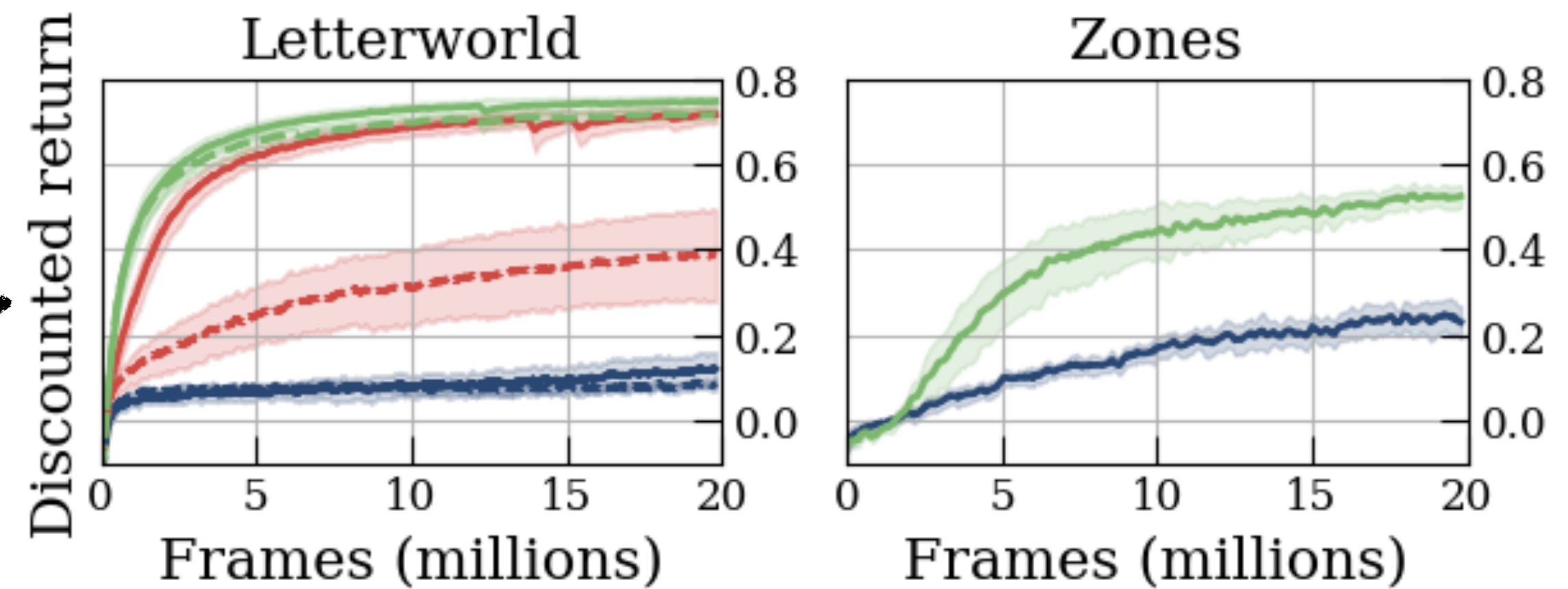
Yalçınkaya, Beyazıt, et al. 'Compositional embedding for natural-condition reinforcement learning' Neurips 2024

Yalcinkaya, Beyazit, et al. "Autanta-Conditional Cooperative Multi-Agent Reinforcement Learning." ML 2026.

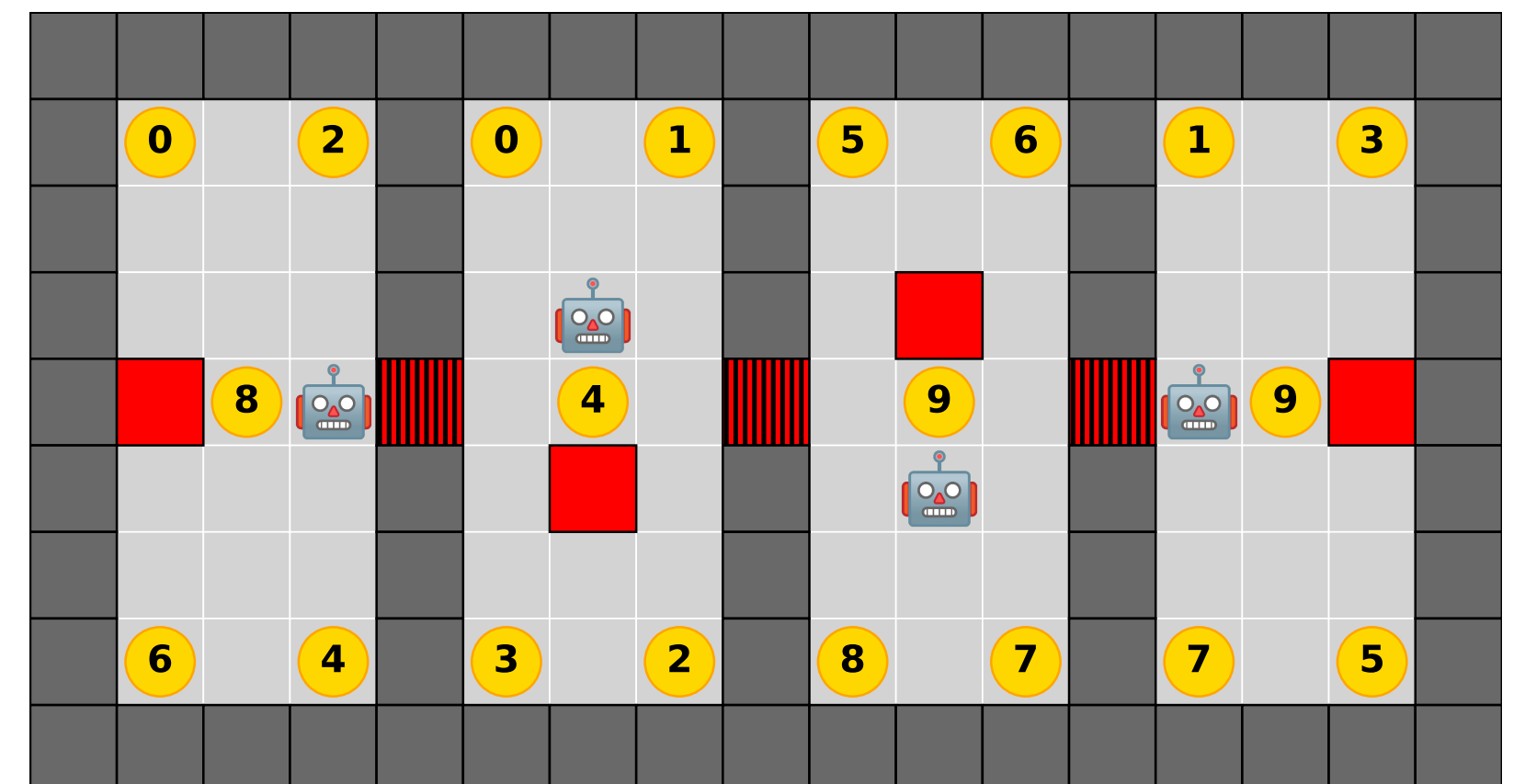
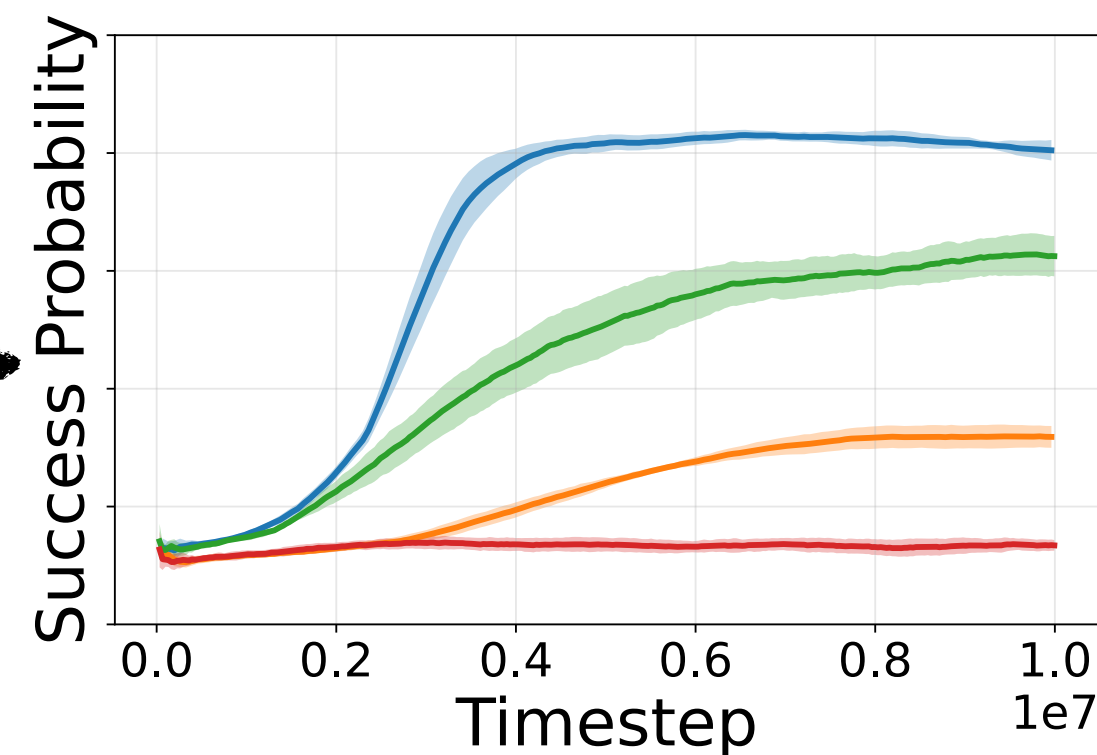
RAD Embeddings

- Improve sample efficiency in TC-RL and TC-MARL.

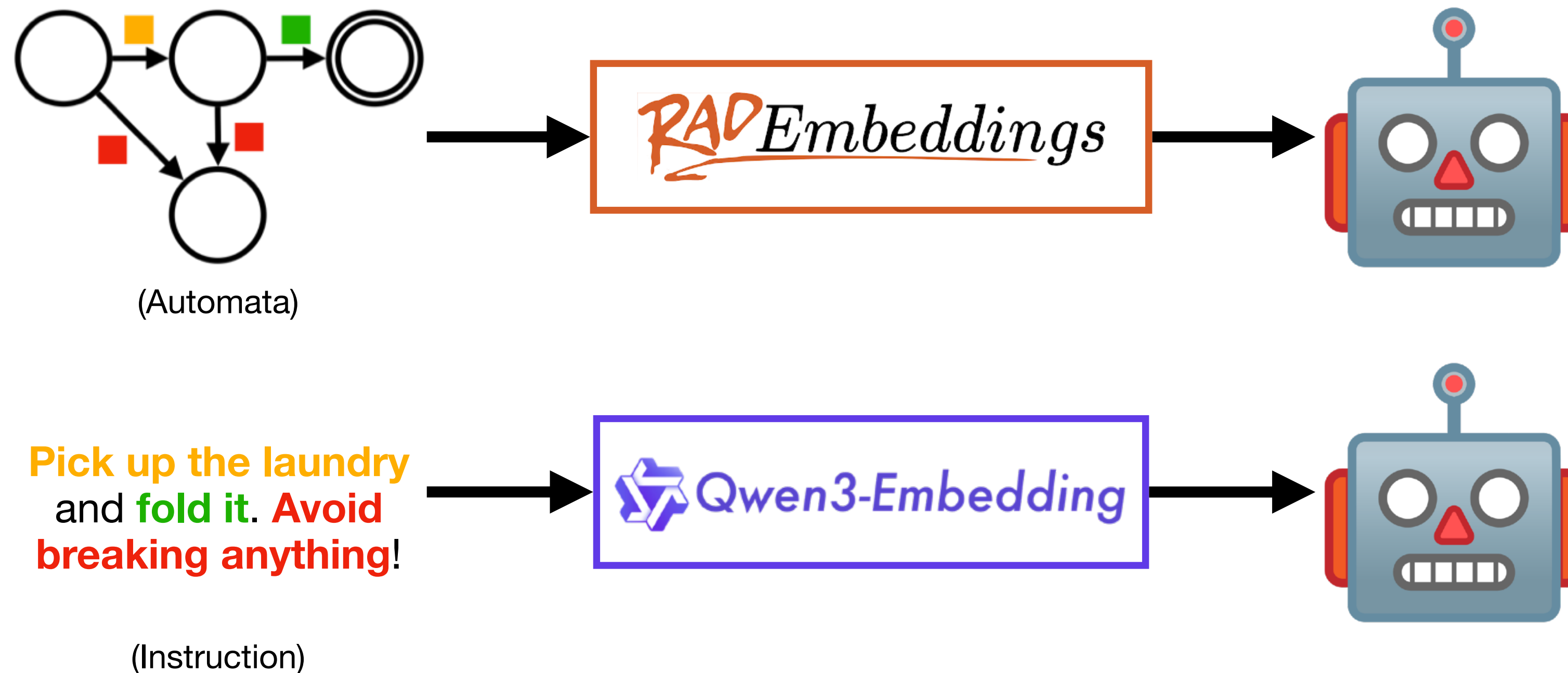
**Automata-Conditioned
RL**



**Automata-Conditioned
Cooperative MARL**



Recap: Our goal is to compare these two modalities.



1. Which modality is better for learning-based control?
 2. What happens when these modalities are combined?
- **Challenge:** How to generate automaton-instruction pairs?